Federated Execution Manager

A Generic Solution for Federated Analysis in Biomedical Projects

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Federated Analytics
Analysis of distributed datasets without moving data. Aggregated statistics or patterns locally computed at each site.

Federated Learning
Collaborative machine learning where models are trained across decentralized data without sharing the raw data.

(Data) Federation
A decentralized architecture where each data site retains control over its data, computation, and privacy policies.

EUCAIM Project
European Cancer Imaging Initiative supporting the creation of a federated infrastructure to share cancer imaging and AI tools across borders.

AI4HI Projects Contributions

INCISIVE · ProCAncer-I · EUCAN IMAGE · PRIMAGE (Medical imaging Artificial intelligence Childhood cancer research) · chAIMELEON

Clinical Contributions

The EUropean Federation for CAncer IMages (EUCAIM) brings together 76 partners across Europe. It launched with 23 clinical sites, including 19 major university hospitals from 12 countries, and aims to expand to 30+ distributed data providers across 15 countries by the end 2026.

What is the Federated Execution Manager (FEM)?

FEM is a lightweight orchestrator for federated tasks.

- ➔ Coordinating data workflows across sites.
- ➔ Managing federated training and analytics.
- ➔ Handling task distribution, role assignment, and result collation.
- ➔ Interfacing with 3rd-party tools (e.g., FL frameworks, Docker, Kubernetes).

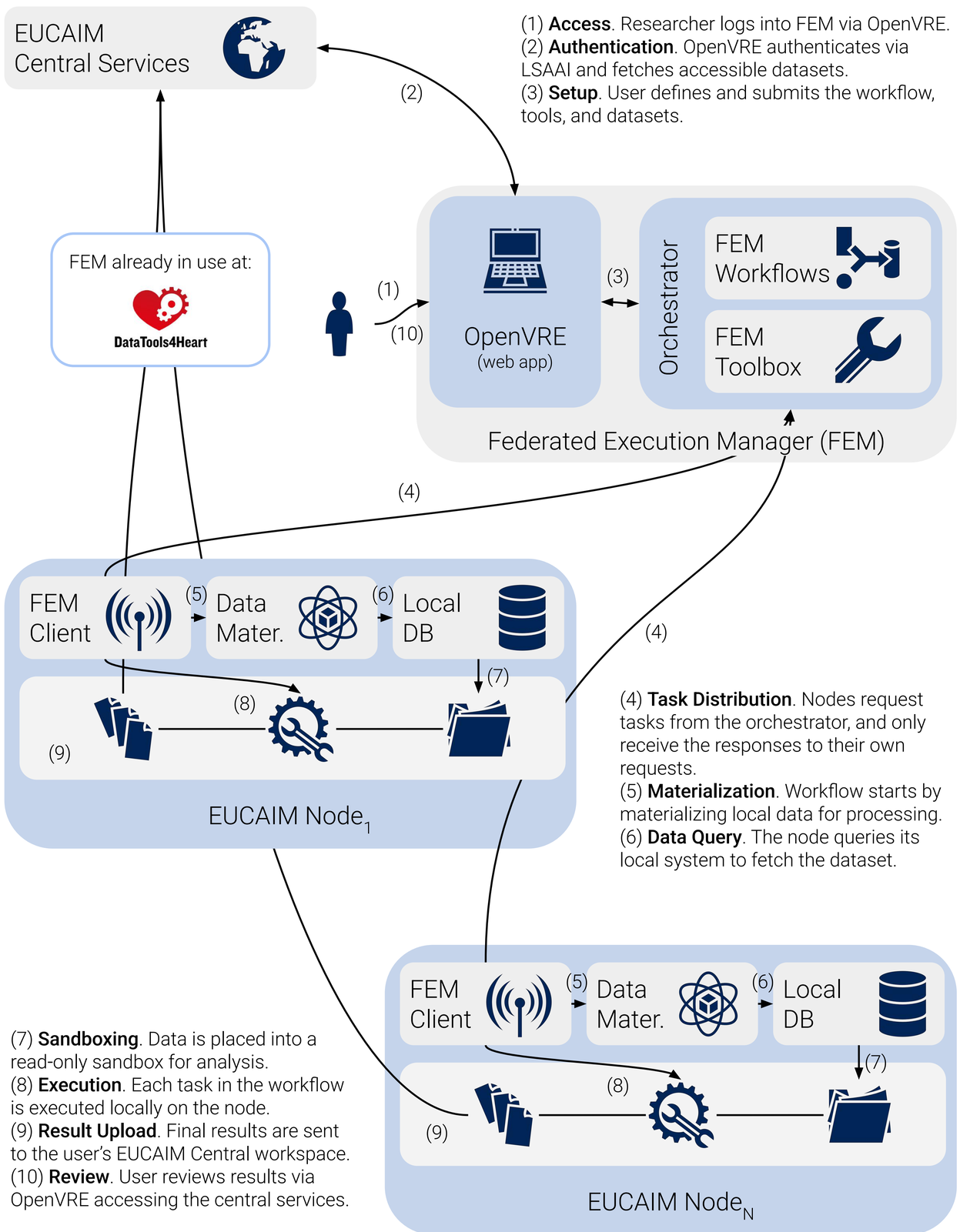
FEM's Main Features

- ➔ **Modular Architecture**. Atomic bundles for transformation analytics, learning, tasks (using containers).
- ➔ **Workflows**. Using Federated CWL (an adapted version of CWL).
- ➔ **Federated Role Handling**. Assign roles (e.g. server/client) per task.
- ➔ **Runtime Agnostic**. Works with Docker, K8s, or custom scripts.
- ➔ **Metadata-Rich**. Supports tracing, compliance, and reproducibility.

FEM Interoperability

The manager interfaces with:

- ➔ **Node Heterogeneity**. Data, compute, mixed nodes.
- ➔ **FL Frameworks Compatible**. Such as Flower or Fed-BioMed among others.
- ➔ **Execution-platforms Independent**. Can work with docker, k8s, SLURM, etc.



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Support

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